

Getting the next generation to learn the GNU/Linux system

James “Jim” Ed Randson (jimedrand)

jimedrand@disroot.org



**Before we're going to the
topic, let me introduce
myself for a first time.**

Speaker profile

I am called James Ed Randson, but I am also known as Jim. The author is employed on a part-time basis as a software developer and package maintainer within a home repository within openSUSE Build Service. I primarily utilise Linux as my operating system of choice, and I am a dedicated member of the Linux community. For a period of time, the author pursued a career as a web developer. Currently, I am enrolled at the Indonesia Open University (Universitas Terbuka), where I am pursuing the undergraduate Information Systems programme.

The extracurricular activities engaged in by the subject are not of a singularly distinctive nature. The subject's engagement in such activities as viewing Japanese animation, documentaries, listening to AM/SW radio, and watching television is indicative of a conventional lifestyle; however the subject does engage in a number of other activities. The author's academic interest lies primarily in engineering, with a particular emphasis on computing and associated fields. Further insights into my academic and professional background can be found in the biography I have authored, accessible via <https://en.opensuse.org/User:Jimed-rand>.





openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

Introducing GNU/Linux

Learn the history first



openSUSE.
Asia Summit 2026
Yogyakarta, October 3 - 4



What is GNU/Linux?

GNU/Linux, in its capacity as an operating system, is comprised of a series of foundational software applications, which facilitate a number of functions within a computing environment. These include the interpretation of user instructions; the reading and writing of data onto and from storage devices and printers; the management of memory; and the execution of other software applications. Of particular note is the Linux kernel, which constitutes the most critical component of the system under discussion. In this particular system, Linux functions as the kernel, with the majority of other components originating from the GNU Project. It is important to note that the Linux kernel, in and of itself, does not constitute a complete operating system. Consequently, the term "GNU/Linux" has been deemed to be more accurate than the popular designation "Linux". Designed according to the Unix model as a multitasking system, it is notable for being developed by a community of volunteers rather than by a single entity, as is the case with many free and open-source software projects. The development of GNU began in 1984 when the Free Software Foundation launched the GNU Project, with the objective of producing free software for a wide range of tasks, including file manipulation, document editing, and programming. The project rapidly became a significant contributor to software development, philosophy, and community.

The initial release of the Linux kernel occurred in 1991, when Linus Torvalds, a Finnish computer science student, unveiled an early iteration of a kernel designed to supersede Minix via the Usenet newsgroup comp.os.minix. To this day, Torvalds coordinates hundreds of developers with the assistance of various subsystem maintainers. From a user perspective, GNU/Linux is an operating system that offers the freedom to choose software. This includes the option of selecting from a variety of command-line shells and graphical desktop environments. The system is regarded as offering enhanced stability, the capacity to operate multiple programs simultaneously, and superior security in comparison to a number of alternative operating systems. This configuration of the operating system has been identified as a major contributing factor to its accelerated development within the server market. Moreover, it has recently garnered heightened levels of popularity among residential and corporate users.



You see the tree in your left? That's how distribution works. Actually, the distribution is kind of a complete, ready-to-use operating system created by bundling the Linux kernel with GNU software tools, system libraries, and application packages. openSUSE is one of them, and most people would know like Debian, Ubuntu (Debian-based), Fedora, etc. They distribute the different goods like any rules, which package manager, which init system, and more.

**GNU delivers the
software, while Linux
itself is the kernel that
GNU used it**

Challenge that People faces when they look at GNU/Linux

Like “Lock-In vendor”, technical issue, not pre-loaded, “cannot open the app that most people use”, etc

Lock-in Vendor, The Most Problems that People have never realised about it

Lock-in Vendor is actually the problem where people got to use the software that they think is needed and it gives you the UX that people will always remember and might be facing the skepticism when the alternatives is available.

The example would be like when people already take the lesson to make the document, spreadsheet, or even presentation, once they know the software that they're looking for (for this case, like MS365), they will only focusing on the software that they know and use it (or we called "got trapped in hellhole"). When they see the another software that did use the different format, they sometimes got panicked like "how to use it" or sometimes it rejected to use alternatives.

Technical Issues

This one is also one of other issues that most people facing, the technical issues. When people see the computer, the OS that they would like to know is only one (or two if we include modern Apple computers), like *Microsoft Windows* and *Apple macOS* (on Apple Mac devices). And also, not anyone could be able to fix the problem without asking anyone on the forum (or even asking the LLM).

The other thing besides that issue is the **Alienation** when they see the Unix-like systems. They like think it's only for server or for "weird" people in their "have no idea" minds. Not only Alienation, but also they facing another issues to understand how GNU/Linux works.

Lack of Literation and “The Limitation”

“The Limitation” here is similar to previous slide that I explained, but actual limitation here is not only the user’s interest. GNU/Linux is free and open-source software, and people could do anything they want to control their own computer. So, when those companies knows user’s right to control anything on their own, they panicked and create some shield like “AntiCheat” on the game or DRM-ed goods that Linux user cannot access it.

Not only that, but it also happenes because of lack of literation. For those people who have interested so much and they are interested, they hunger of information about the software they wanna use. If we gave them 80/20 or 90/10 dependencies policy that 80 or 90 of them are adapting the free and open-source software, and the rest of them are less to touch the proprietary software.

Another challenge that user faces

It's not only those three that I mentioned on previous slide, it is so much challenge that anyone faces when they see the OS that they never seen it before.

Another challenge that user faces

It's not only those three that I mentioned on previous slide, it is so much challenge that anyone faces when they see the OS that they never seen it before.

**You have big right to control you
computers and no one should limit it**

Why Microsoft Windows is bad?

Microsoft Windows in these days is really bad and annoying. Not only they're proprietary, they also because forced system updates, privacy-invading data collection, and intrusive advertisement. Forced update is really annoying on most time that people use it. Also, the consumer edition is really bloated and "full of junk".

The Performance is Improving better when running GNU/Linux rather than Microsoft Windows

The GNU/Linux operating system has been demonstrated to exhibit superior performance benchmarks when compared to the Microsoft Windows operating system. This superior performance is attributable to the GNU/Linux system having a significantly reduced background footprint. While a conventional Windows installation typically operates with a memory allocation of between 2 and 8 GB, often due to background telemetry, mandatory software updates, and pre-installed software, numerous prominent Linux distributions operate with comparable efficacy on less than 1 GB of memory. This considerable diminution in resource overhead engenders the availability of a substantially augmented amount of hardware power for the facilitation of user applications. Furthermore, the Linux system is demonstrably superior in its administration of memory and the execution of input/output operations, resulting in expedited file extraction, accelerated code compilation processes, and significantly diminished application launch times.

The performance benefits of this approach are most evident when utilising older hardware that exhibits challenges in processing the demands of contemporary Windows operating systems, or when executing intensive software development workflows. In the context of contemporary PC gaming, compatibility layers such as Valve's Proton facilitate the attainment of near-identical or occasionally superior framerates between Linux and Windows systems, a phenomenon that is especially salient in configurations employing AMD graphics hardware. In essence, by removing superfluous background processes, the Linux operating system ensures that the processors and memory of the computer are dedicated exclusively to executing the desired tasks.

You handle your computers

Basically, the title means using GNU/Linux does have benefitically good to use it both for personal use and highend enterprise use. And it gives you to touch the actual system like root access, and you can control how it should be like, etc. You own the thing that runs on your own computer hardware.

Cracking Software is not always good to use

Some people who could've still using cracked software still has no idea why are they running those and they don't even know what's the big risks behind it. So, if you already using Linux, you can take a few time to learn of using alternative software, rather than keeping to use the cracked software that it is not very good to use.

**How I can get the good part of using
GNU/Linux and try to reduce the
unnecessary goods?**



If you still scared, try to install VM

VM software like VMware (Workstation, Fusion, EsXI), Microsoft (Hyper-V), Oracle (VirtualBox), QEMU/KVM, and UTM (QEMU on Mac) could be a helper solution for you if you still scared at first to try it. For the distros, go check distrowatch.com or you can try any distros that you'd like. If you're interested to use openSUSE, then go to <https://get.opensuse.org> and download it.

Get out from the Hell holes

Hell holes in this context means that you're still trapped at "Vendor Lock-in" and if you care about sovereignty on your own, you can get out of there and try to adapt with FOSS ecosystems. Even if you sometimes still use proprietary software, you can apply the 80/20 or 90/10 dependencies policy. I'll be explain it on next slide.

80/20 or 90/10 dependencies policy.

This dependencies policy makes you to setting up the 80 or 90 into your rate of using FOSS ecosystem and few remaining means you still depends on it in few things. Also, this policy is effective if you can control what do you want to use and how many times that you want to avoid unnecessary goods in there. The example it looks like when you are working on Audio engineering, you can take the big point on your main control, and the lower point which is the proprietary points, you can use it as second point.

And many other things

You can learn from experienced Linux users beside myself, and you can apply it many other things that people have done, and you will get the good results in the end.

Wanna reach more about me?

You can find me on the Internet
under my public username
“jimedrand” on search engines and
you can contact me anytime when
you’re interested to discuss
anything with me. I am opening my
gate to anybody.



Thank You
Matur Nuwun
Terima Kasih



openSUSE®
Asia Summit 2026